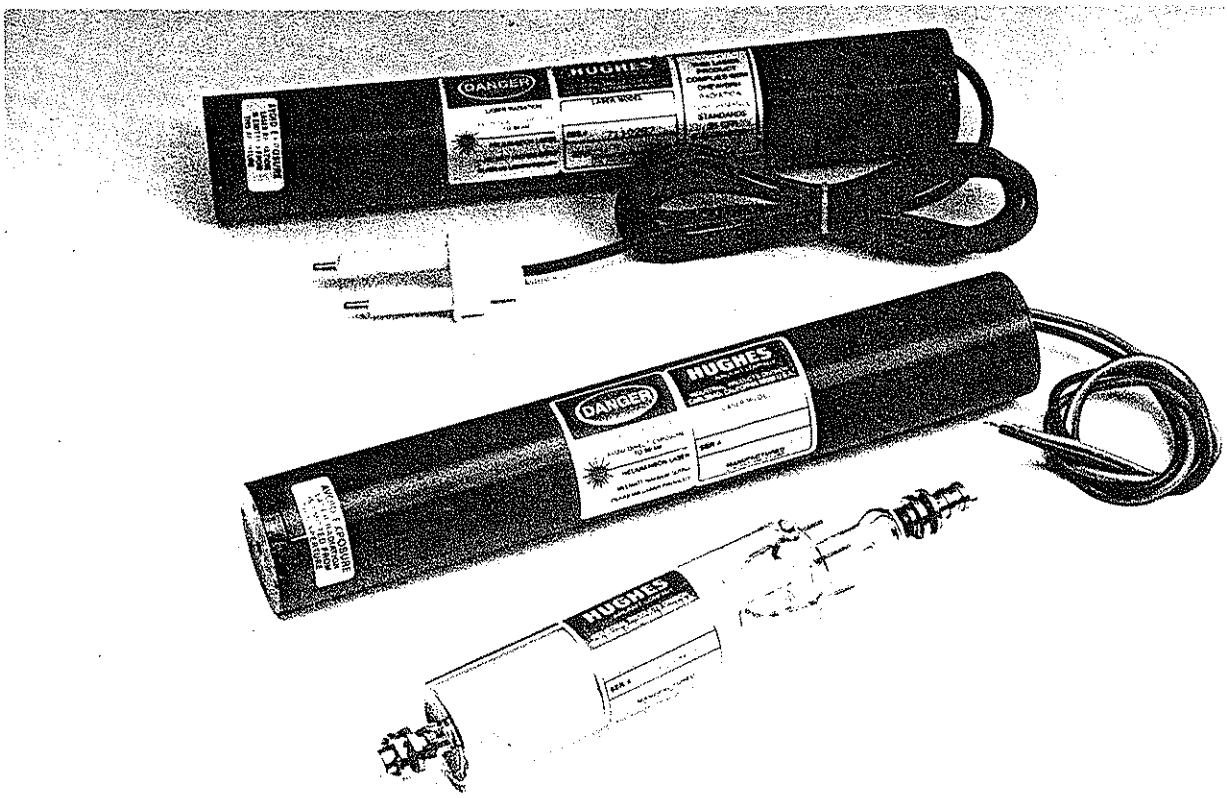
HUGHES
AIRCRAFT COMPANY




# HUGHES LF Series
# 1 and 2 mW Helium-Neon Lasers

***PROVEN OEM PERFORMANCE***

- Hughes hard-seal construction
- Rugged coaxial internal mirror construction
- Specially developed hard glass

OEMs depend on the LF Series' stable performance, long trouble-free operation and high reliability.

OEMs depend on Hughes world reputation for leading the way in lasers.

Hughes LF Series lasers are backed by a 12-month warranty (18 months available for qualified OEMs). Specifications are subject to change without notice.

# LASERS SPECIFICATIONS

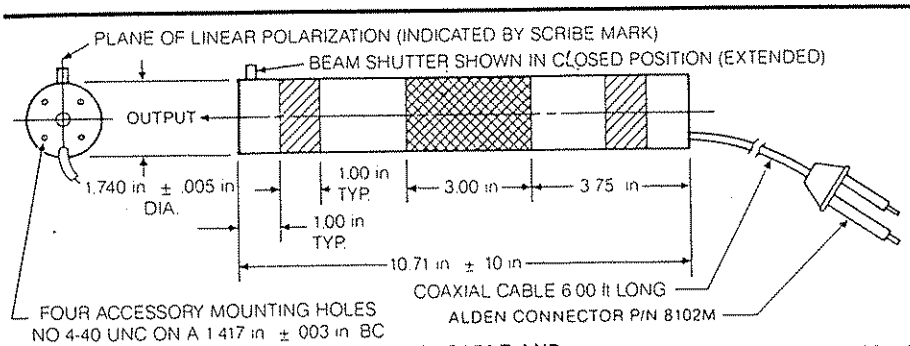
| | LASER HEAD MODEL NO. | | | LASER PLASMA TUBE MODEL NO. | |
|---|---|---|---|---|---|
| | 3221H<br>3221H-C<br>3221H-P<br>3221H-PC | Flying Leads<br>High Voltage Conn.<br>Polarized/Flying Leads<br>Polarized/High Voltage Conn. | 3222H<br>3222H-C<br>3222H-P<br>3222H-PC | 3121H<br>3121H-P | 3122H<br>3122H-P |
| • Minimum output power $TEM_{00}$ at 632.8 nm (mW) | 1.0 | | 2.0 | 1.0 | 2.0 |
| • Beam diameter $1/e^2$ (mm) | 0.64 | | 0.64 | 0.64 | 0.64 |
| • Beam divergence (mrad) | 1.3 | | 1.3 | 1.3 | 1.3 |
| • Longitudinal mode spacing (c/2L)(MHz) | 685 | | 685 | 685 | 685 |
| • Operating voltage (Vdc ± 100) | 1740 | | 1740 | 1350 | 1350 |
| • Recommended series anode ballast (kΩ) | Included in head | | Included in head | 60 to 75 | 60 to 75 |
| • Connection with power source | Flying leads or high voltage connector Alden No. 8102M | | Flying leads or high voltage connector Alden No. 8102M | — | — |

## COMMON SPECIFICATIONS

- Amplitude noise (30 Hz-10 Mhz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drift ± 5% (in any 8 hr. period following 15 min. warm-up)
- Starting voltage < 10.0 kVdc
- Optimum operating current 6.5 ± 0.2 mA
- Static alignment (head) centered to outer cylinder to ≤ 0.010 in, parallel to outer cylinder to ≤ 1 mrad
- Angular drift < 0.2 mrad from cold start at 25°C < 0.03 mrad after 15-minute warm-up
- Polarization ("P" models only) — Linear, 500:1 extinction ratio

## COMMON ENVIRONMENTAL SPECIFICATIONS

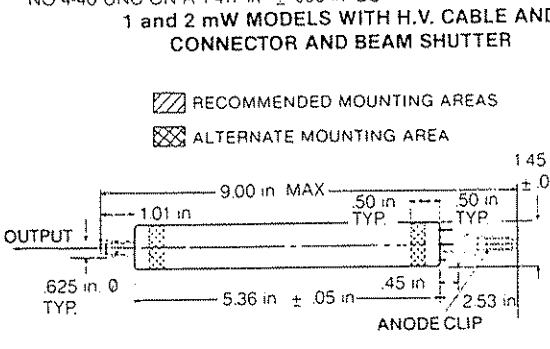
| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | – 20°C to 50°C | – 40°C to 80°C |
| Altitude | 0-10,000 ft. | 0-70,000 ft. |
| Relative humidity without condensation | 0-100% | 0-100% |
| Shock | | 50 g for 1 ms<br>15 g for 11 ms |



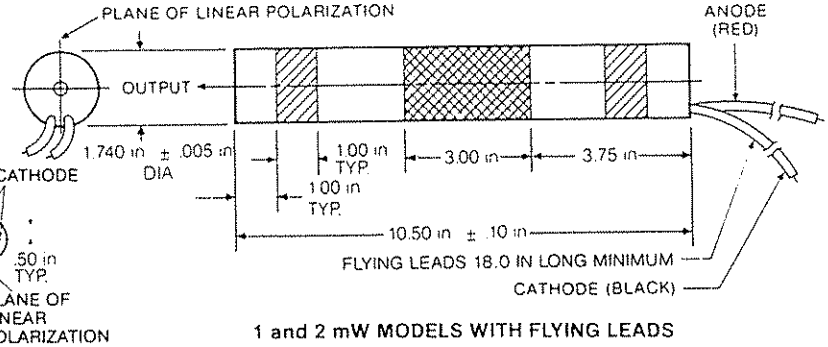

1 and 2 mW MODELS WITH H.V. CABLE AND CONNECTOR AND BEAM SHUTTER




3221H, H-C, H-P, H-PC
3222H, H-C, H-P, H-PC




INTERNAL MIRROR PLASMA TUBE




1 and 2 mW MODELS WITH FLYING LEADS

*Specifications subject to change without notice*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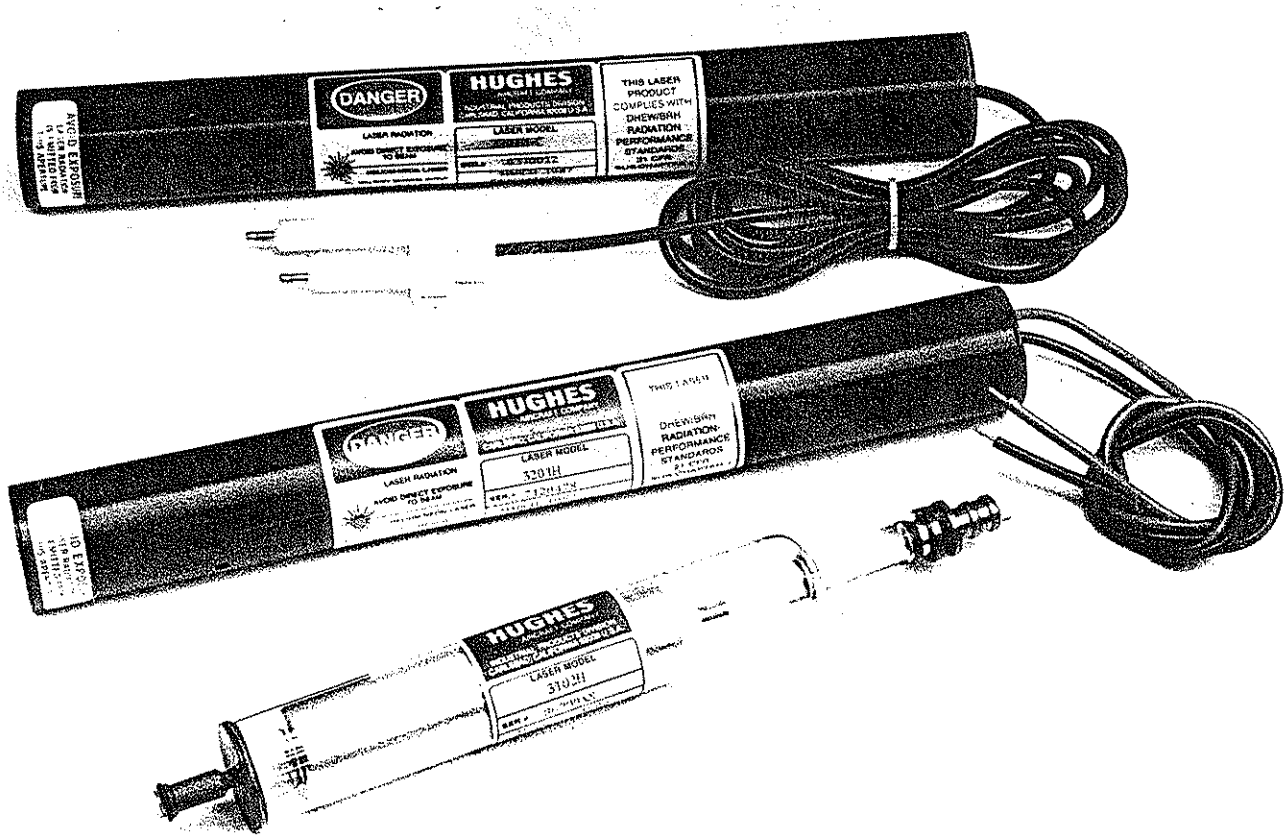



# HUGHES LC Series
# 1 and 2 mW Helium-Neon Lasers

***PROVEN OEM PERFORMANCE***

The LC Series are compact lasers featuring hard seal construction and a patented cathode design. This gives OEM users the best performance and lifetimes possible.

Hughes LC Series lasers are backed by a 12-month warranty (18 months available to qualified OEMs). Specifications are subject to change without notice.

# LASERS SPECIFICATIONS

| | LASER HEAD MODEL NO. | | | LASER PLASMA TUBE MODEL NO. | |
|---|---|---|---|---|---|
| | 3201H<br>3201-C<br>3201H-P<br>3201H-PC | Flying Leads<br>High Voltage Conn.<br>Polarized/Flying Leads<br>Polarized/High Voltage Conn. | 3203H<br>3203H-C<br>3203H-P<br>3203H-PC | 3101H<br>3101H-P | 3102H<br>3102H-P |
| • Minimum output power $TEM_{oo}$ at 632.8 nm (mW) | 1.0 | | Random 2.0<br>Polarized 1.5 | 1.0 | Random 2.0<br>Polarized 1.5 |
| • Beam Diameter $1/_e2$ (mm) | 0.49 | | 0.49 | 0.49 | 0.49 |
| • Beam Divergence (mrad) | 1.7 | | 1.7 | 1.7 | 1.7 |
| • Longitudinal mode spacing (c/2L)(MHz) | 635 | | 635 | 635 | 635 |
| • Operating voltage (Vdc ± 100) | 1770 | | 1770 | 1430 | 1430 |
| • Required series anode ballast (kΩ) (Ballast location 3 inches from anode.) | Included in head | | Included in head | 75 | 75 |
| • Connection with power source | Flying leads or high voltage connector Alden No. 8102M | | Flying leads or high voltage connector Alden No. 8102M | — | — |

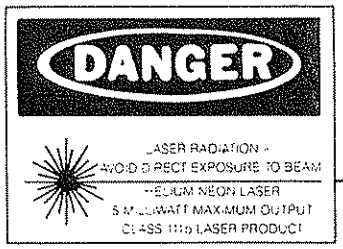
## COMMON SPECIFICATIONS

- Amplitude noise (30 Hz-10 MHz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drive ± 5% (in any 8 hr. period following 15 min. warm-up)
- Starting voltage < 8 kVdc
- Optimum operating current 4.5 ± 0.2 mA
- Static alignment (head) centered to outer cylinder to ≤ 0.010 in., parallel to outer cylinder to ≤ 1 mrad
- Angular drift
  < 0.2 mrad from cold start at 25 °C
  < 0.03 mrad after 15-minute warm-up
- Polarization Liner, 500:1
  ("P" models only) extinction ratio

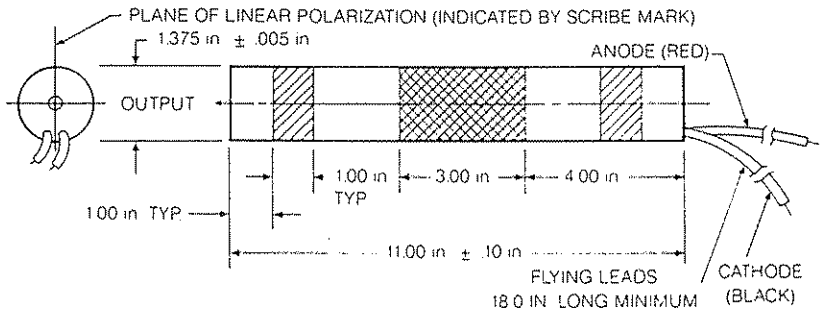
## COMMON ENVIRONMENTAL SPECIFICATIONS

| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | –20 °C to 50 °C | –40 °C to 80 °C |
| Altitude | 0-10,000 ft | 0-70,000 ft |
| Relative humidity without condensation | 0-100% | 0-100% |
| Shock | | 50 g for 1 ms<br>15 g for 11 ms |

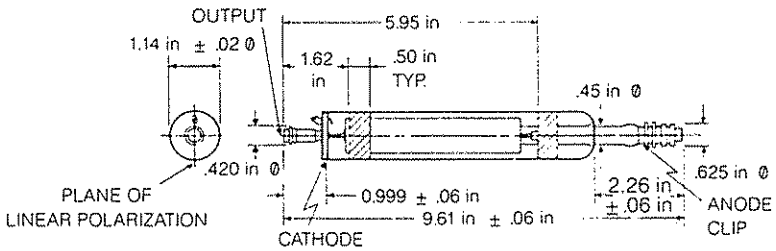
*Specifications subject to change without notice.*



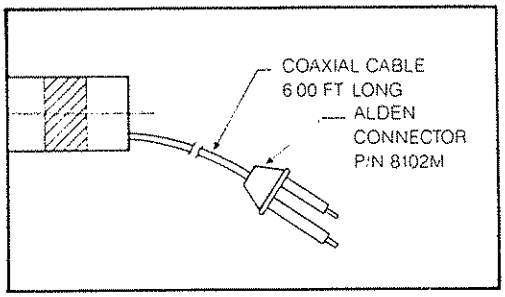

3201H, H-C, H-P, H-PC
3203H, H-C, H-P, H-PC




MODELS WITH FLYING LEADS ( H AND H-P MODELS)




INTERNAL MIRROR PLASMA TUBE




WITH H.V. CONNECTOR (H-C and H-PC MODELS)
(All other dimension same as H and H-P models)

RECOMMENDED MOUNTING AREAS
ALTERNATE MOUNTING AREA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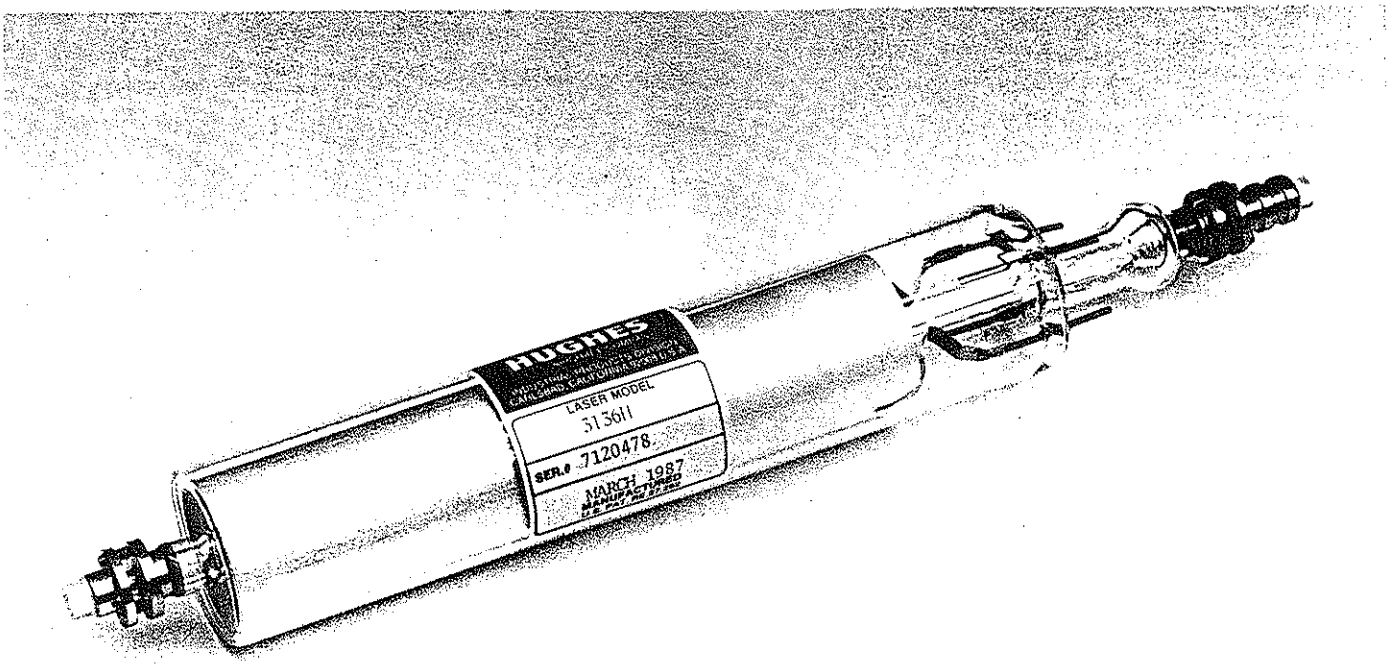



# HUGHES LF Series
# Model 3136H
# 2 mW Helium-Neon Laser

***PROVEN OEM PERFORMANCE***

- Hughes hard-seal construction
- Rugged coaxial internal mirror construction
- Specially developed hard glass
- Hard-seal replacement for the 80-2T

OEMs depend on the LF Series' stable performance, long trouble-free operation and high reliability.

Hughes LF Series lasers are backed by a 12-month warranty (18 months for qualified OEMs).

# LASER SPECIFICATIONS

## INTERNAL MIRROR PLASMA TUBE MODEL NO. 3136H

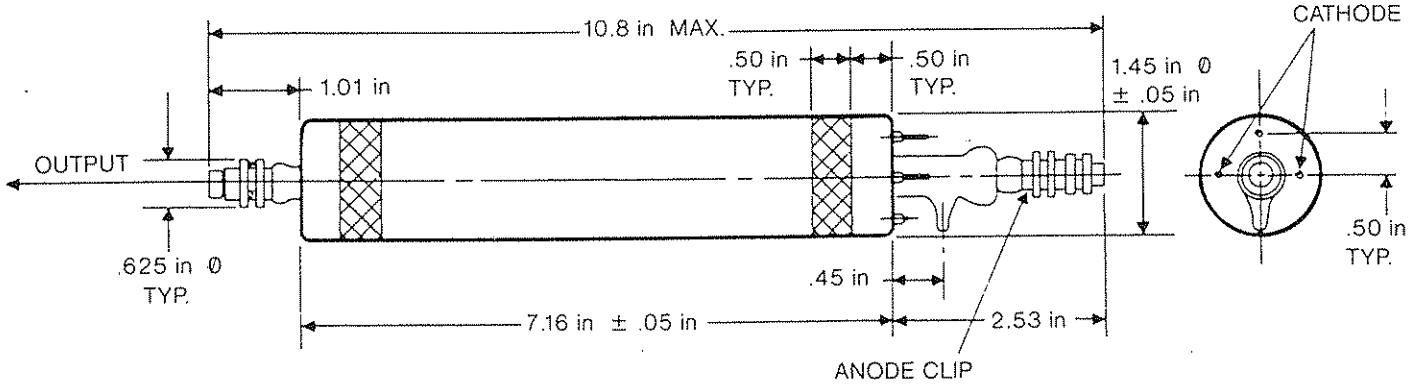
| | |
|---|---|
| • Minimum output power $TEM_{oo}$ at 632.8 nm (mW) | 2.0 |
| • Beam diameter $1/_e2$ (mm) | .83 |
| • Beam divergence (mrad) | 1.0 |
| • Longitudinal mode spacing (c/2L) (MHz) | 575 |
| • Operating voltage (Vdc ± 100) | 1450 |
| • Recommended series anode ballast (kΩ ) | 60 to 75 |

## COMMON SPECIFICATIONS

- Amplitude noise (30 Hz-10 MHz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drive ± 5% (in any 8-hour period following 15-minute warm-up)
- Starting voltage < 10.0 kVdc
- Optimum operating current 6.0 ± 0.2 mA
- Angular drive <0.2 mrad from cold start at 25°C <0.03 mrad after 15-minute warm-up

## COMMON ENVIRONMENTAL SPECIFICATIONS

| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | –20°C to 50°C | –40°C to 80°C |
| Altitude | 0-10,000 ft. | 0-70,000 ft. |
| Relative humidity without condensation | 0-100% | 0-100% |
| Shock | | 50 g for 1 ms<br>15 g for 11 ms |




RECOMMENDED MOUNTING AREAS

## CENTER FOR DEVICES AND RADIOLOGICAL HEALTH

Laser models on this data sheet are sold only on an OEM or export basis and have not been certified relative to CDRH performance standard 21CFR1040. It is the user's responsibility to certify product compliance where applicable.

*Specifications are subject to change without notice*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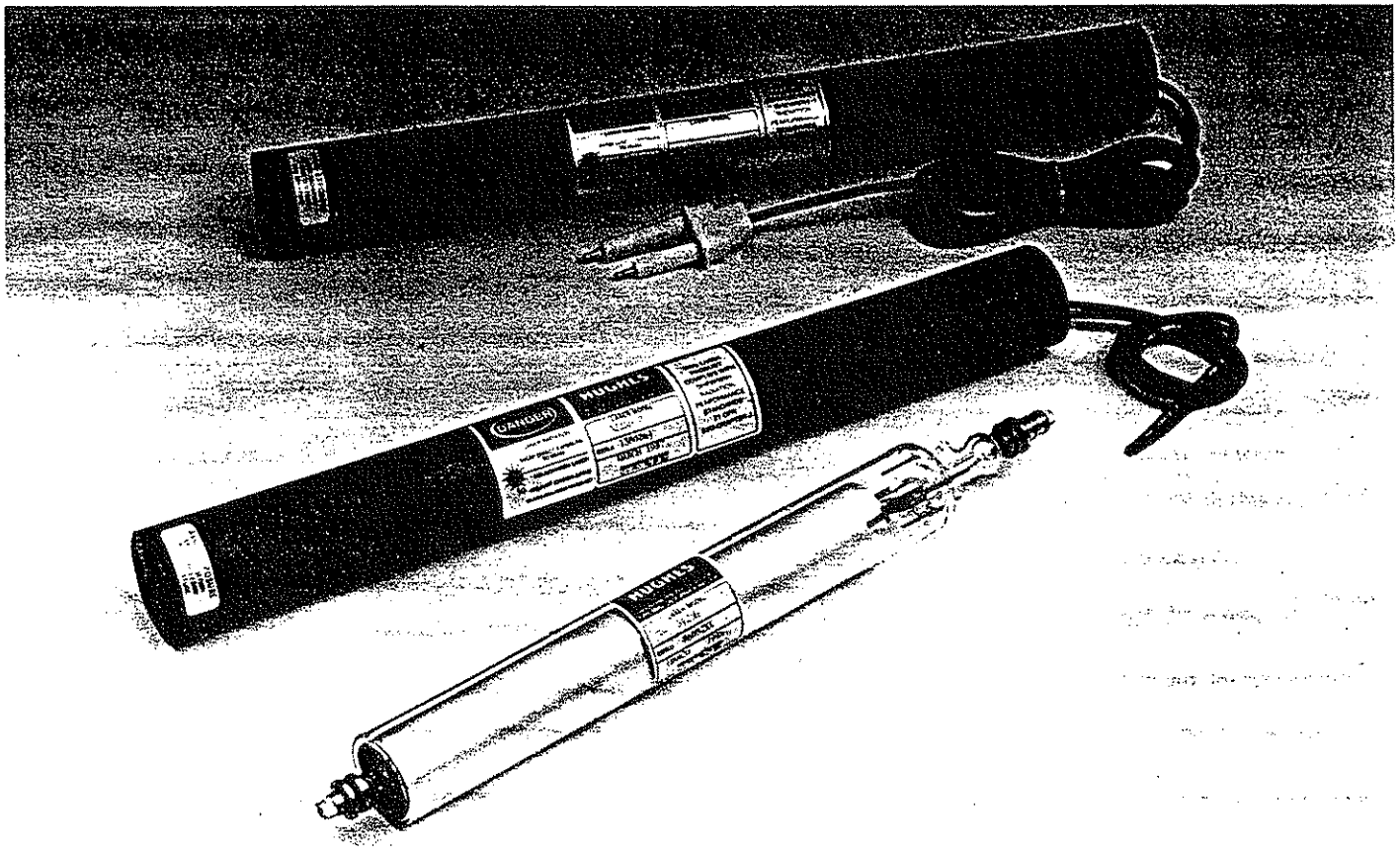


# HUGHES LF Series
# 4 and 5 mW Helium-Neon Lasers

***PROVEN OEM PERFORMANCE***

- Hughes hard-seal construction
- Rugged coaxial internal mirror construction
- Specially developed hard glass

OEMs depend on the LF Series' stable performance, long trouble-free operation and high reliability.

OEMs depend on Hughes world reputation for leading the way in lasers.

Hughes LF Series lasers are backed by a 12-month warranty (18 months available for qualified OEMs). Specifications are subject to change without notice.

# LASER SPECIFICATIONS

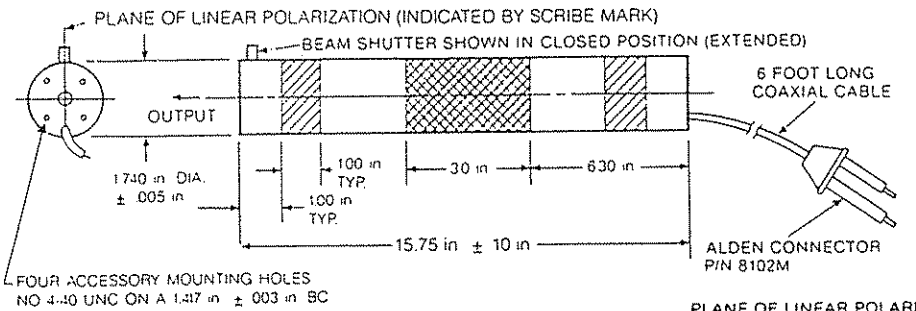
| | LASER HEAD MODEL NO. | | LASER PLASMA TUBE MODEL NO. | |
|---|---|---|---|---|
| | 3224H (Flying Leads) | 3225H | 3124H | 3125H |
| | 3224H-C (High Voltage Conn.) | 3225H-C | 3124H-C | 3125H-C |
| | 3224H-P (Polarized/Flying Leads) | 3225H-P | 3124H-P | 3125H-P |
| | 3224H-PC (Polarized/High Voltage Conn.) | 3225H-PC | | |
| • Minimum output power $TEM_{00}$ at 632.8 nm (mW) | 4.0 | 5.0 | 4.0 | 5.0 |
| • Beam diameter $1/e^2$ (mm) | 0.83 | 0.83 | 0.83 | 0.83 |
| • Beam divergence (mrad) | 1.0 | 1.0 | 1.0 | 1.0 |
| • Longitudinal mode spacing (c/2L)(MHz) | 435 | 435 | 435 | 435 |
| • Operating voltage (Vdc ± 100) | 2300 | 2300 | 1910 | 1910 |
| • Recommended series anode ballast (kΩ) | Included in head | Included in head | 60 to 75 | 60 to 75 |
| • Connection with power source | Flying leads or high voltage connector Alden No. 8102M | Flying leads or high voltage connector Alden No. 8102M | — | — |

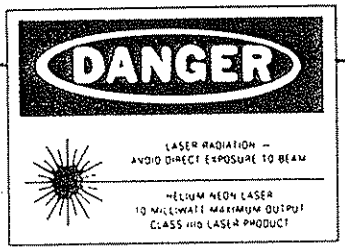
## COMMON SPECIFICATIONS

- Amplitude noise (30 Hz-10 Mhz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drift ± 5% (in any 8 hr. period following 15 min. warm-up)
- Starting voltage ≤ 10.0 kVdc
- Optimum operating current 6.5 ± 0.2 mA
- Angular drift < 0.3 mrad from cold start at 25°C. < 0.04 mrad after 15-minute warm-up
- Static alignment (head) centered to outer cylinder to ≤ 0.010 in. parallel to outer cylinder to ≤ 1 mrad
- Polarization ("P" models only) — Linear, 500:1 extinction ratio

## COMMON ENVIRONMENTAL SPECIFICATIONS

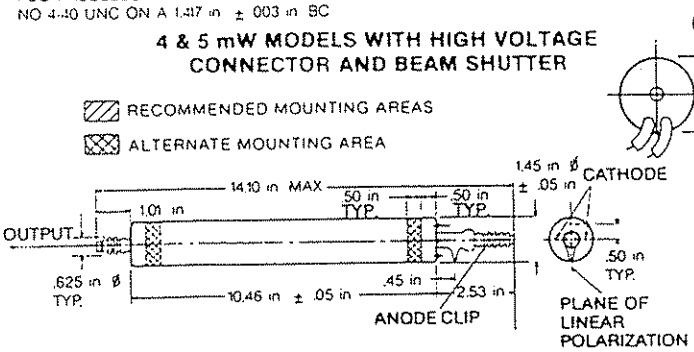
| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | – 20°C to 50°C | – 40°C to 80°C |
| Altitude | 0-10,000 ft | 0-70,000 ft |
| Relative humidity without condensation | 0-100% | 0-100% |
| Shock | | 50 g for 1 ms<br>15 g for 11 ms |



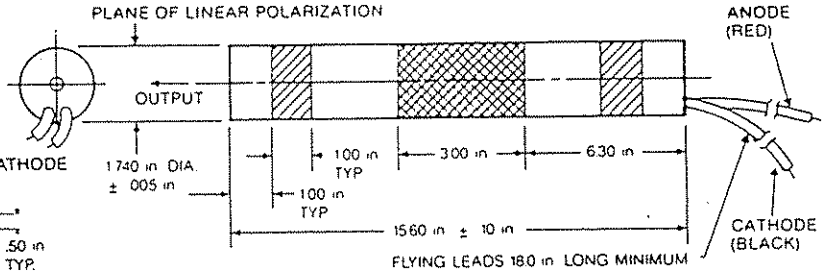




3224H, H-C, H-P, H-PC
3225H, H-C, H-P, H-PC

**4 & 5 mW MODELS WITH HIGH VOLTAGE CONNECTOR AND BEAM SHUTTER**




**INTERNAL MIRROR PLASMA TUBE**




**4 & 5 mW MODELS WITH FLYING LEADS**

*Specifications subject to change without notice*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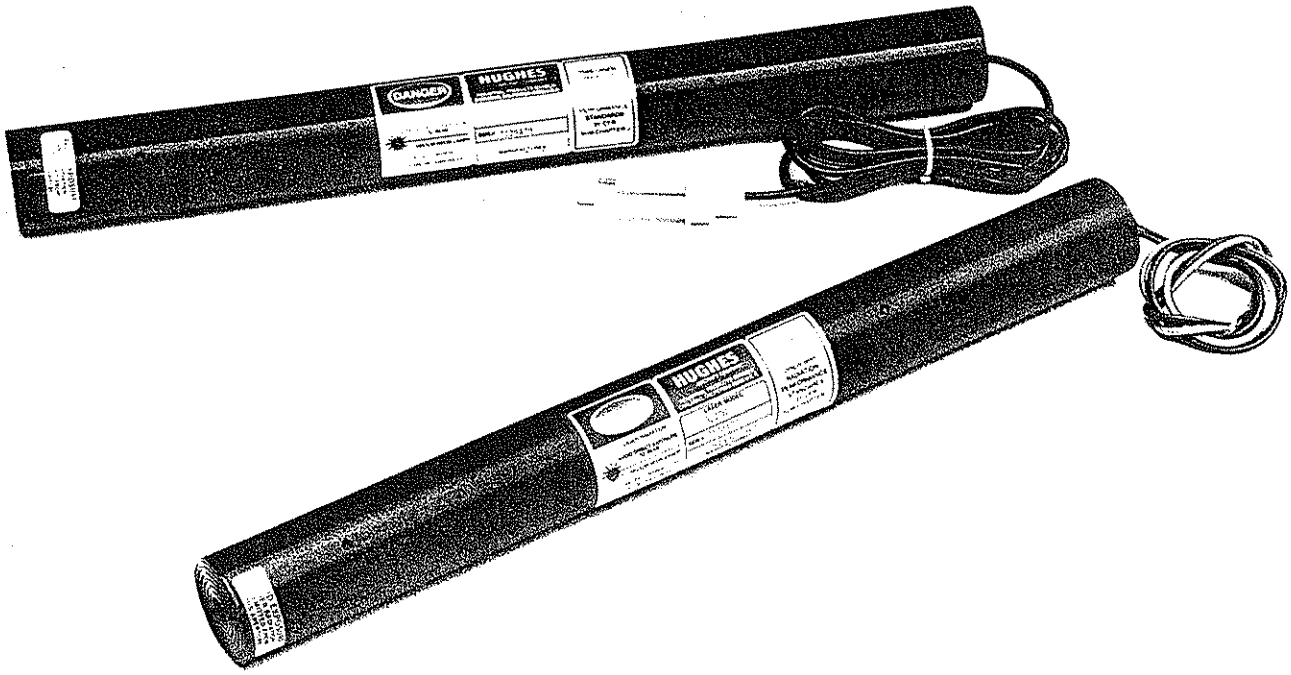



# HUGHES LF Series
# 7.0 mW Helium-Neon Lasers

***PROVEN OEM PERFORMANCE***

- Hughes hard-seal construction
- Rugged coaxial internal mirror construction
- Specially developed hard glass

OEMs depend on the LF Series' stable performance, long trouble-free operation and high reliability.

OEMs depend on Hughes world reputation for leading the way in lasers.

Hughes LF Series lasers are backed by a 12-month warranty (18 months available for qualified OEMs). Specifications are subject to change without notice.

# LASER SPECIFICATIONS

| LASER HEAD MODEL NO. | |
|---|---|
| High Voltage Conn. | 3227H-C |
| Flying Leads | 3227H |
| Polarized/Flying Leads | 3227H-P |
| Polarized/High Voltage Conn. | 3227H-PC |

| | |
|---|---|
| • Minimum output power $TEM_{00}$ at 632.8 nm (mW) | 7.0 |
| • Beam diameter $1/_e2$ (mm) | .82 mm |
| • Beam divergence (mrad) | 1.1 |
| • Longitudinal mode spacing (c/2L)(MHz) | 410 |
| • Operating voltage (Vdc ± 100) | 2600 |
| • Recommended series anode ballast (kΩ) | Included in head |
| • Connection with power source | High voltage connector Alden No. 8102M |

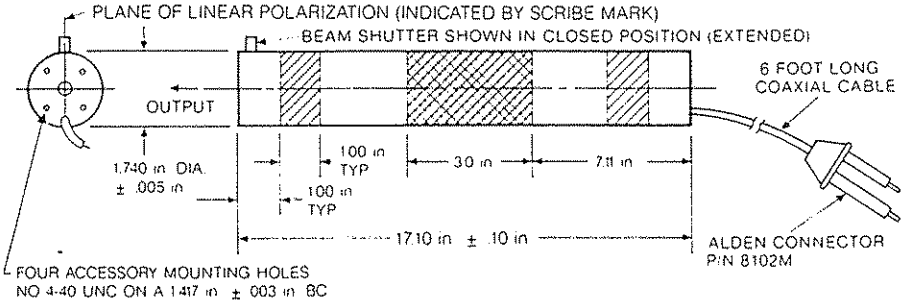
## COMMON SPECIFICATIONS

- Amplitude noise (30 Hz-10 Mhz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drift ± 5% (in any 8 hr. period following 15 min. warm-up)
- Starting voltage ≤ 10.0 kVdc
- Optimum operating current 7.0 ± 0.2mA
- Angular drift < 0.3 mrad from cold start at 25°C, < 0.04 mrad after 15-minute warm-up
- Static alignment (head) centered to outer cylinder to ≤ 0.010 in, parallel to outer cylinder to ≤ 1 mrad
- Polarization ("P" models only) — Linear, 500:1 extinction ratio

## COMMON ENVIRONMENTAL SPECIFICATIONS

| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | – 20°C to 50°C | – 40°C to 80°C |
| Altitude | 0-10,000 ft | 0-70,000 ft |
| Relative humidity without condensation | 0-100% | 0-100% |
| Shock | | 15 g for 11 ms |

3227H, H-C, H-P, H-PC




**7.0 mW MODELS WITH HIGH VOLTAGE CONNECTOR AND BEAM SHUTTER**

▨ RECOMMENDED MOUNTING AREAS

▨ ALTERNATE MOUNTING AREA

*Specifications are subject to change without notice*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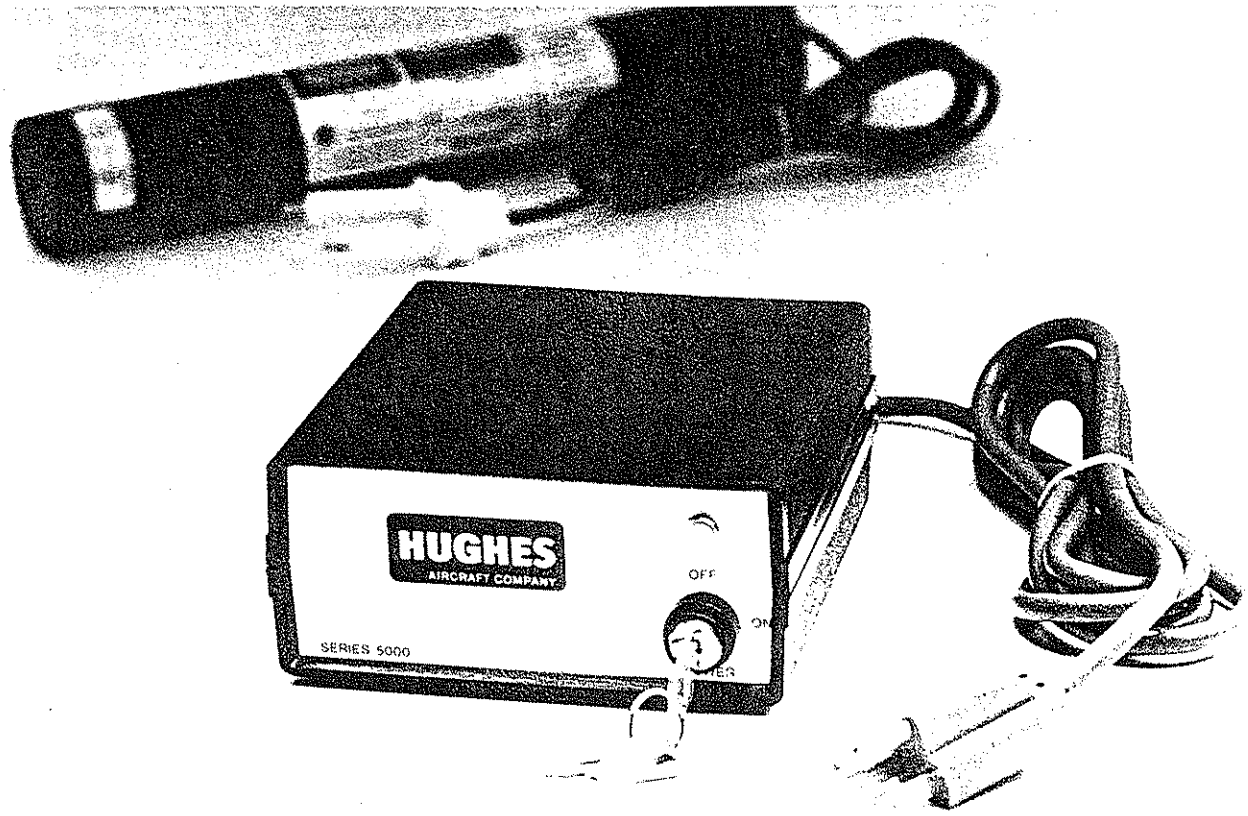



# HUGHES 5000 Series

# 115/230 Vac CDRH Certified Helium-Neon Laser Power Supply

***OPERATES 0.5 mW to 7.0 mW LASER HEADS***

- Ultra compact configuration
- Contains all CDRH required accessories for Class II and IIIb laser products
  - — key activated switch
  - — ignition time delay
  - — pilot light emission indicator
  - — remote interlock connector

## GENERAL DESCRIPTION

The 5000 series power supplies have been designed to provide optimum electrical interface with Hughes 0.5 through 7 mW laser heads. The high power conversion-efficient design, coupled with feedback regulated starting voltage and output current regulation circuits, ensure maximum laser performance and system reliability. The patented electronic current filtering technique yields a laboratory quality dc output current with an ultra low ripple component. Packaged in a rugged, compact enclosure, the 5000 series power supplies are suitable for both OEM and laboratory applications.

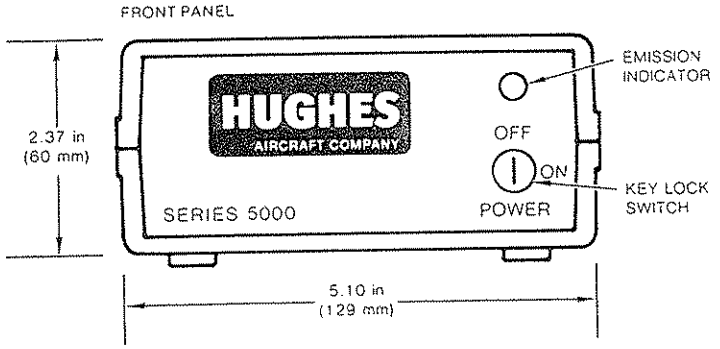
## POWER SUPPLY MODELS

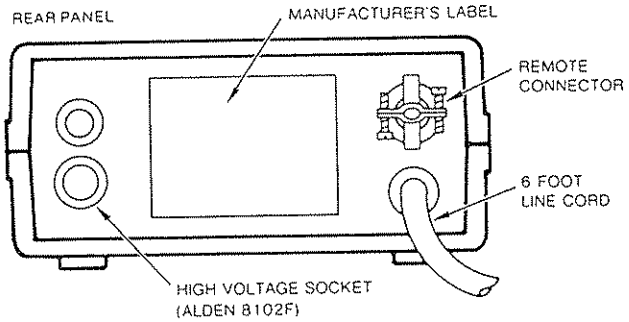
| | 5000 | 5010 | 5020 | 5040 |
|---|---|---|---|---|
| For operation with laser model | 3209H-C<br>3209H-PC | 3223H-C<br>3223H-PC | 3221H-C 3221H-PC<br>3222H-C 3222H-PC<br>3224H-C 3224H-PC<br>3225H-C 3225H-PC | 3227H-C<br>3227H-PC |
| Output start voltage (feedback regulated) (kVdc min) | 8 | 10 | 10 | 10 |
| Output sustaining voltage (Vdc) | 1550-1750 | 1550-1750 | 1630-2450 | 2600-2800 |
| Laser operating current (mA±0.05) | 4.5 | 7.0 | 6.5 | 7.0 |
| Laser beam amplitude ripple (PTP) (Includes only the laser beam amplitude ripple component attributable to the power supply.) | <0.5% | <0.5% | <0.5% | <1.0% |

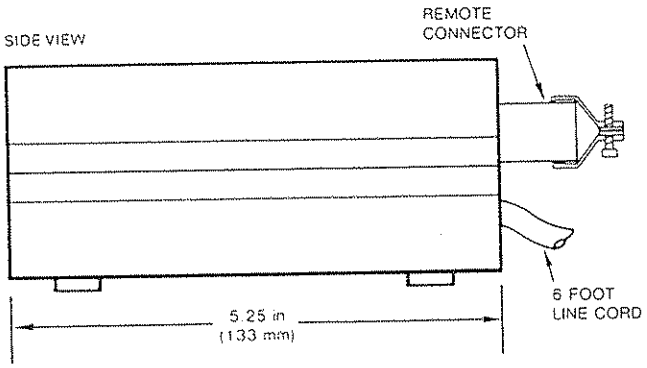
## COMMON SPECIFICATIONS

- Ignition delay — 3-5 s nominal
- Mass — 2.3 lb/1.1 kg
- Input power — 115 Vac ± 10% or 230* Vac ± 10% (50-400 Hz)
- Operating temperature — −20°C to +50°C

*To order models for 230 Vac operation, add the letter "F".




*Specifications are subject to change without notice*

**HUGHES AIRCRAFT COMPANY • Industrial Products Division**
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**HUGHES AIRCRAFT SYSTEMS INTERNATIONAL**
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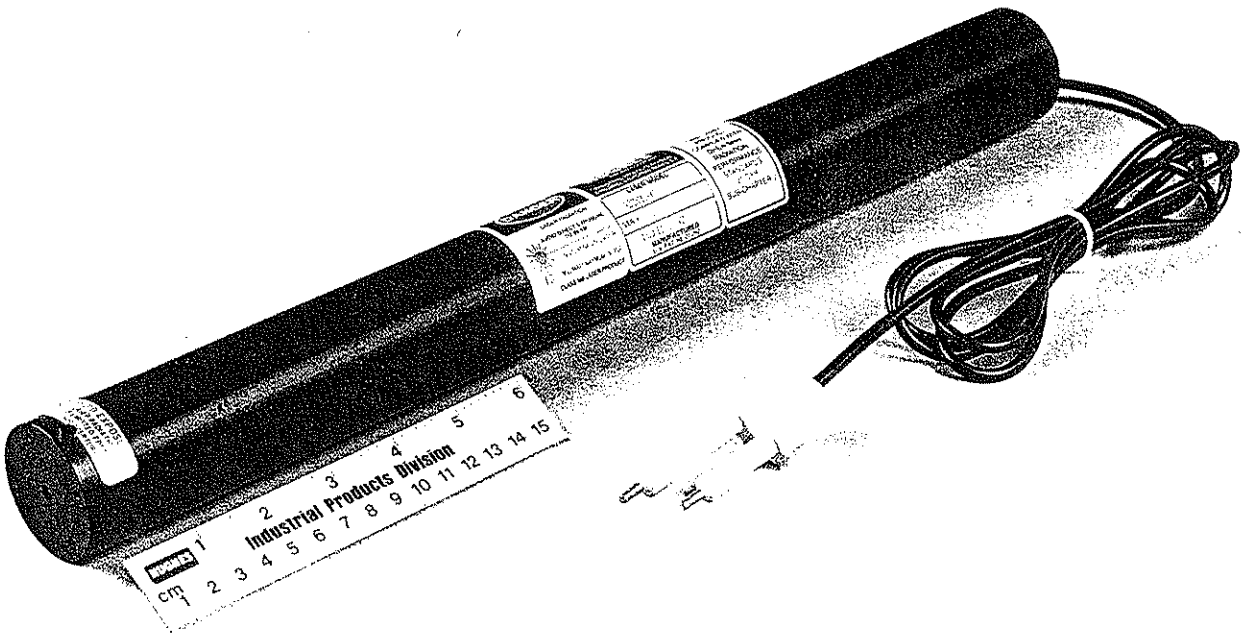
HUGHES

Subsidiary of GM Hughes Electronics




# HUGHES LF Series
# 10.0 mW Helium-Neon Lasers

- Hughes hardseal construction
- Rugged coaxial internal mirror construction
- Specially developed hard glass

***PROVEN OEM PERFORMANCE***

OEMs depend on the LF Series stable performance, long trouble-free operation and high reliability.

OEMs depend on Hughes' world reputation for leading the way in lasers.

Hughes LF Series lasers are backed by a 6-month warranty. Specifications are subject to change without notice.

# LASER SPECIFICATIONS

| LASER HEAD | MODEL NO. |
|---|---|
| High Voltage Connector | 3230H-C |
| Polarized/High Voltage Connector | 3230H-PC |

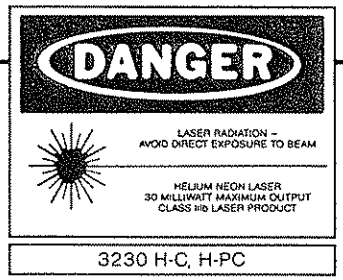
| | |
|---|---|
| • Minimum output power $TEM_{00}$ at 632.8 nm (mW) | 10.0 |
| • Beam diameter $1/e2$ (mm) | .68 mm |
| • Beam divergence (mrad) | 1.2 |
| • Longitudinal mode spacing (c/2L)(MHz) | 350 |
| • Operating voltage (Vdc ± 100) | 2800 |
| • Recommended series anode ballast (kΩ) | Included in head (60 kΩ) |
| • Connection with power source | High voltage connector Alden No. 8102M |

## COMMON SPECIFICATIONS

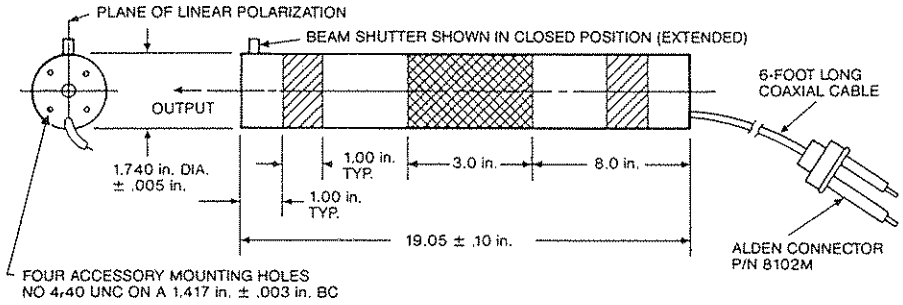
- Amplitude noise (30 Hz–10 MHz) < 1% rms
  Note: Ripple is generally higher with dc power supply
- Long term drift ± 5% (in any 8-hour period following 15-minute warm-up)
- Starting voltage ≤ 10.0 kVdc
- Optimum operating current 7.0 ± 0.2mA
- Angular drift < 0.3 mrad from cold start at 25°C < 0.04 mrad after 15-minute warm-up
- Static alignment (head) centered to outer cylinder to ≤ 0.010 inch, parallel to outer cylinder to ≤ 1 mrad
- Polarization ("P" models only) — Linear, ≥ 500:1 extinction ratio

## COMMON ENVIRONMENTAL SPECIFICATIONS

| | OPERATING | NON-OPERATING |
|---|---|---|
| Temperature | – 20°C to 50°C | – 40°C to 80°C |
| Altitude | 0–10,000 ft | 0–70,000 ft |
| Relative humidity without condensation | 0–100% | 0–100% |
| Shock | | 15 g for 11 ms |



3230 H-C, H-PC




**10.0 mW MODELS WITH HIGH VOLTAGE CONNECTOR AND BEAM SHUTTER**

RECOMMENDED MOUNTING AREAS

ALTERNATE MOUNTING AREA

*Specifications are subject to change without notice*

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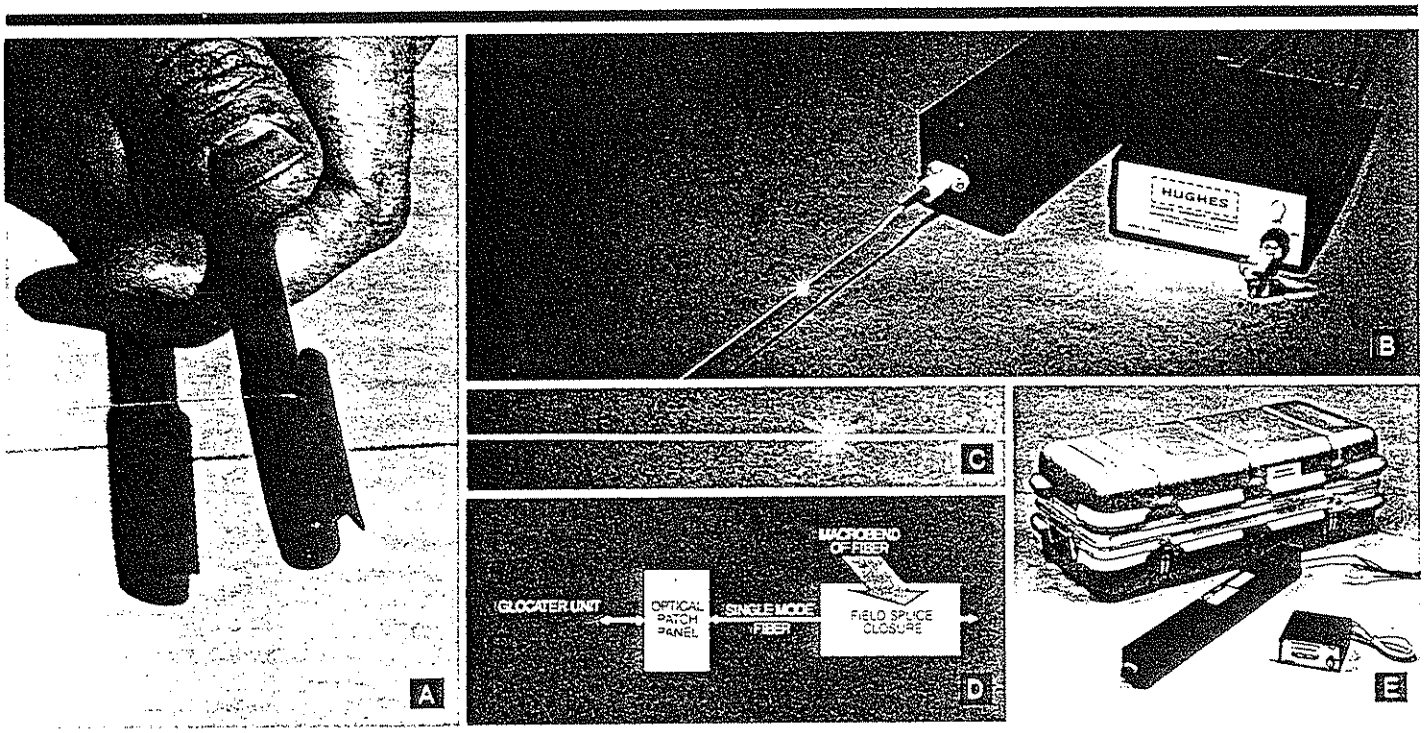
# Glocater™
# Fiberoptic Fault Locater

MELLES GRIOT




*Easily induced macrobend (12 mm is ideal) produces visible glow (A). Breaks show clearly (B) and (C). Connects directly to patch panel (D). Comes complete with carrying case (E).*

## A totally new concept for spotting fiberoptic faults instantly

Now, you can visually inspect optical fiber continuity during installation, maintenance and other procedures. This new fiberoptic fault locater identifies fibers, fiber breaks or imperfections using visible laser light.

The unique tool—consisting of a helium-neon laser, power supply and high efficiency fiber coupler—is simple to use. Just attach the unit to fibers under test, and the coherent red light emitted from the laser produces a bright orange glow at fault locations or bends. The glow is so intense that it is easily seen at breaks or imperfections through the outer coatings of fiber jumpers, pigtails or buffer tubes.

For fiber identification, the laser's visible red light appears in continuous fiber after inducing a constant macrobend. Existing fiber breaks and macrobends will produce a glow even without handling the fiber.

The Glocater fault locater is designed for use in the installation, rearrangement and maintenance of fiberoptic cables as well as in system acceptance testing of cables, jumpers and pigtails. It can also serve as an excellent tool for fiberoptic training centers. It can be used on single mode or multimode fiber of any wavelength—for example, 850, 1300 or 1550 nanometers—to verify continuity and positive end-to-end identification.

### TYPICAL APPLICATIONS

Locating breaks. Connect the Glocater fault locater at the interconnect cabinet. Breaks in the pigtails, jumpers or coated fiber will emit an orange glow that is visible directly through the outer coating or buffer tubes anywhere within the cabinet.

Identifying fibers. Attach the fault locater to a fiber end. Bend the fiber over a finger or marking pen at any point along the fiber route. The orange glow appears at the bend to positively identify the fiber.

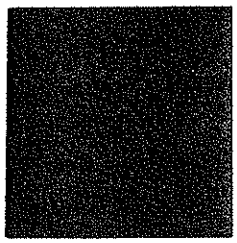
Locating points of attenuation. Connect the fault locater to a fiber end. At violated fiber bends, the glow can be seen through the jacketing on pigtails, jumpers or stored fiber.

The Glocater fiberoptic fault locater is compact, lightweight and comes complete with carrying case. It's an easy, affordable way to track down fiberoptic problems. Call or write today for full information or to request a demonstration.

NEW! Now available with FC connector

RELIABLE • He-Ne laser for stable performance
ECONOMICAL • Advanced fault detection at an affordable price
EASY TO OPERATE • Simple connections are all that's needed
SAFE TO USE • Built-in interlock for safe operation
COMPACT • Carrying case holds complete unit

# GLOCATER FIBEROPTIC FAULT LOCATOR SPECIFICATIONS



| | |
|---|---|
| Transmitter | |
| Wavelength | 632.8 nm |
| Output level | +5 dBm or more (10/125 μm single mode) |
| Visible range*, nominal | |
| Single mode, one way | 3.0 miles or more at –32 dBm |
| Multimode, one way | 2.5 miles or more at –50 dBm |
| Power requirements | 115/220 Vac 50/60 Hz |
| Connectors | Choice of biconic or fiber connector (FC); specify<br>Adaptable to other connectors (consult factory) |
| Dimensions (HxWxL) | |
| Transmitter | 2.6x2.5x24.5 in (67x64x622 mm) |
| Power supply | 2.4x5.1x5.3 in (60x129x133 mm) |
| Weight | |
| Transmitter | 4.4 lb (2.0 kg) |
| Power supply | 2.3 lb (1.1 kg) |
| Accessories supplied | Power supply equipped with safety interlock and power cord<br>Carrying case |

*Visible red light observed after inducing a constant macrobend of 12 mm in diameter.

*Specifications subject to change without notice.*

*Customer service assistance in evaluating proposed applications is available.*

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